



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (24/08/2026 – 30/08/2026)

WEEK 5

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

**DESIGN AND IMPLEMENT GRAPHICAL USER
INTERFACE FOR AN OPEN SOURCE FIREWALL**

GROUP MEMBER :

PRATYUSH ANAND, ADITYA SINGH

GROUP ID :

2026G306

GROUP GUIDE :

VEDANT JOSHI

GROUP COORDINATOR :

SIDHDHARTH PATEL

COLLEGE GUIDE NAME :

EHT SHAM

COLLEGE GUIDE No. :

+91 9818938592

COLLEGE GUIDE EMAIL :

Ehte.sham@pilani.bits-pilani.ac.in

COLLEGE NAME :

BITS PILANI

24/08/2026	Reviewed previous prototype output and identified that ChromaDB's default distance metric was producing invalid negative similarity percentages for lower ranked retrieval results.
25/08/2026	Fixed the retrieval scoring by switching ChromaDB to cosine distance and reran the prototype to confirm all similarity scores now fall in a valid range with no negative values.
26/08/2026	Holiday (Onam or Thiru Onam Day)
27/08/2026	Expanded the document corpus from 3 to 5 government circulars (dummy)(added MKSY irrigation scheme and Swachh Bharat Gramin sanitation scheme) tested semantic retrieval with different queries and confirmed match confidence improved across all sample questions.
28/08/2026	Holiday (Rakhshabandhan)
29/08/2026	Documented before/after results comparing retrieval confidence pre and post-fix and confirmed the metric fix and corpus expansion together roughly doubled match confidence on average (e.g. 41.7%→70.8%, 7.9%→54.0%).
30/08/2026	Holiday (Sunday)

Next Week Plan	Evaluate the prototype across different query types, then begin with the structured (SQL) data prototype.
-----------------------	---

REFERENCE:

<https://docs.trychroma.com/docs/overview/introduction>

<https://huggingface.co/sentence-transformers>

<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

<https://www.youtube.com/watch?v=QojA5urlDzA>

<https://www.youtube.com/watch?v=JfSmffOyV-8>

SCREEN SHOTS:

```
def main():
    model = SentenceTransformer("all-MiniLM-L6-v2")
    print("      Done - 384-dimensional local embeddings ready.")

    # 2. Set up ChromaDB
    print("\n[2/4] Initializing ChromaDB...")
    client = chromadb.Client()
    collection_name = "government_circulars"
    try:
        client.delete_collection(collection_name)
    except Exception:
        pass
    # Use cosine distance instead of Chroma's default squared-L2 distance.
    # Squared-L2 can exceed 1, which made "1 - distance" produce negative
    # "similarity" percentages in the previous version. Cosine distance
    # stays in a 0-2 range, so 1 - distance is a meaningful similarity score.
    collection = client.create_collection(
        collection_name, metadata={"hnsw:space": "cosine"}
    )
    print(f"      Done - collection '{collection_name}' ready (cosine distance).")

    # 3. Load, chunk, embed, and store documents
    print(f"\n[3/4] Loading documents from '{DOCUMENTS_DIR}/'...")
```

```
[1/4] Loading sentence-transformers model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits
and faster downloads.
Loading weights: 100%|████████████████████████████████████████| 103/103 [00:00<00:00, 10939.26it/s]
      Done - 384-dimensional local embeddings ready.

[2/4] Initializing ChromaDB...
      Done - collection 'government_circulars' ready (cosine distance).
```

debug_circular

documents

dlrmp_circular.txt

mksy_circular.txt

pmfby_circular.txt

rera_circular.txt

swachh_bharat_circular.txt

```
[3/4] Loading documents from 'documents/'...  
[OK] Loaded dlrmp_circular.txt -> 2 chunk(s)  
[OK] Loaded mksy_circular.txt -> 2 chunk(s)  
[OK] Loaded pmfby_circular.txt -> 2 chunk(s)  
[OK] Loaded rera_circular.txt -> 2 chunk(s)  
[OK] Loaded swachh_bharat_circular.txt -> 2 chunk(s)  
Total chunks to embed: 10  
Done - embeddings stored in ChromaDB.
```

```
[4/4] Running sample queries...
```

Q1: What is the funding pattern for the Digital Land Records Modernisation Programme?

[1] 70.8% match - dlrmp_circular.txt

Digital Land Records Modernisation Programme (DLRMP) - Circular Objective: The Digital Land Records Modernisation Programme aims to computerize land r...

[2] 35.5% match - rera_circular.txt

Real Estate Regulatory Authority (RERA) - Project Registration Circular Objective: RERA mandates registration of real estate projects to ensure transp...